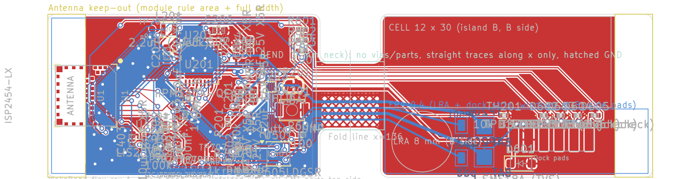
Antenna keep-out (module rule area + full width)
ISP2454-LX
ANTENNA
U1
U201
CELL 12 x 30 (island B, B side)
BEND (9 mm neck): no vias/parts, straight traces along x only, hatched GND
Fold line x=136
LRA 8 mm (B side)
J601
Dock pads
SMF18A (TVS)
Antenna shadow after fold (x 171–176): no copper, no cell, no LRA

WakeBand flex rev A, layout hand-off. Unfolded view, all SMT parts top side.
Pod about 23.5 x 40 mm. Islands 36 x 21.5 mm; the 30 mm cell bay ends 0.5 mm before the antenna shadow.
Stackup: 2-layer adhesiveless PI 25 um, 18 um Cu, coverlay, ENIG, about 0.12 mm.
Stiffeners on the bottom (User.2): FR4 0.4 under island A and under island B's LRA / dock strip
(windows >= 0.3 mm around the B-side pads TP201/202/501/502). No steel stiffeners.
Bend zone (User.3, 9 mm neck): no vias or parts, every trace one straight segment along x, R >= 1.2 mm.
Vias only on stiffened areas, >= 0.5 mm from a stiffener edge. Trace/space >= 0.1 mm.
Confirm with JLCPCB which User layers they read for stiffeners before ordering.

FOLD / ASSEMBLY ORIENTATION (fold line x = 136, island B folds UNDER island A):
- Island A F side (all SMT parts) faces UP, away from the wrist.
- Island B F side faces DOWN to the case floor: dock pads J601 face the dock.
- Cell and LRA mount on island B's B side (inner face, between the islands):
  cell tabs solder to TP201 (+) / TP202 (−), LRA leads to TP501 (+) / TP502 (−), all on B.Cu;
  LRA tapes onto the FR4 stiffener in the 8 mm motor strip beside the cell.
- All SMT parts are on F (single-sided assembly); B-side pads are hand-soldered after the fold.